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Unit – 5 : Session – 3 : SLO – 2 | Graph Traversal — Breadth First Search (BFS)

■ I. Fill in the Blanks

- BFS stands for BREADTH FIRST SEARCH.
- BFS uses QUEUE data structure for its implementation.
- In BFS, a vertex is marked as VISITED when it has been visited.
- The time complexity of BFS for a graph with V vertices and E edges is $O(V + E)$.
- BFS is suitable for finding the SHORTEST path in an unweighted graph.

■ II. True or False

- BFS visits the deepest unvisited node first — FALSE (it visits level by level).
- BFS can be used to detect cycles in an undirected graph — TRUE.
- BFS uses a queue data structure — TRUE.
- In BFS, each vertex is visited exactly once — TRUE.

■ III. Short Answer Questions

- BFS: a traversal that starts at a vertex, visits all its immediate neighbours first, then the neighbours' neighbours, level by level, using a queue.
- Applications: shortest path in unweighted graphs; social-network friend suggestions; also peer-to-peer network search and web crawling.
- Data structure: QUEUE.
- DFS vs BFS: DFS goes deep first using a stack; BFS explores breadth/level-wise using a queue and guarantees shortest hop-count paths in unweighted graphs.

■ IV. Solve and Write — Given Undirected Graph (A-H)

- Adjacency list:
- A -> B, C B -> A, D, E C -> A, F, G D -> B, H
- E -> B, H F -> C, H G -> C, H H -> D, E, F, G
- BFS traversal from A (neighbours in alphabetical order): A, B, C, D, E, F, G, H
- Iterative BFS internally uses a QUEUE.